

Continuous Speech Emotion Recognition

Introduction and Literature Survey

1. Introduction

Speech Emotion Recognition means finding emotion from a person's voice. Usually one audio clip gets one label such as happy, angry or neutral. That works when the clip is short. In a real conversation the mood can change after some time. If we keep only one label for the full audio, we miss when the emotion changed.

If we cut the audio into many small windows and show every prediction, the output becomes messy. One wrong window makes the emotion jump again and again. That is hard for a person to read.

This project does Continuous Speech Emotion Recognition. Continuous here means we look at emotion over time, not only once for the whole file. We predict emotion for each window, remove weak jumps, join same emotions into stable parts, and mark short transition parts. The final answer looks like a timeline, for example: 0-4 sec Neutral, 4-5 sec Transition, 5-8 sec Anger.

This is useful in call centre checks, teaching apps and chatbots where people want to know when the mood changed. Public talk data like MELD already has emotion changes across turns, so we use MELD in this project. Other continuous datasets need college permission and we do not have that.

2. Literature Survey

We studied five related papers. For each paper we write what it does and what it does not cover for our problem.

Paper 1. A Review on Speech Emotion Recognition: A Survey, Recent Advances, Challenges, and the Influence of Noise [1]

This review talks about SER steps like features, classifiers, deep learning, datasets and noise. It gives a clear picture of normal SER work.

Drawbacks:

- Mostly talks about one emotion label for a clip.
- Less talk about emotion changing inside a long conversation.
- Does not show how to make a short timeline from window outputs.

Paper 2. Speech Emotion Recognition Approaches: A Systematic Review [2]

This paper compares many SER methods. It covers features, models and the usual ways papers report scores.

Drawbacks:

- Scores are for fixed segments using accuracy or F1.
- Timeline type output is not discussed.
- Change from one emotion to another is not taken as a separate result.

Paper 3. Speech Emotion Recognition Based on Multi-Feature Speed Rate and Rhythmic Information [3]

This paper uses speaking speed and rhythm with other sound features to get better emotion classification. It shows that how a person speaks matters a lot.

Drawbacks:

- Focus is only on better score for one segment.
- No step to join many window outputs into stable emotion parts.
- No transition marking between emotions.

Paper 4. Speech Emotion Recognition Based on Multi-Dimensional Feature Extraction and Multi-Scale Feature Fusion [4]

This paper takes many speech features and joins them at different scales. This often helps accuracy because short time detail and longer context both are used.

Drawbacks:

- Final answer is still one class for the input segment.
- No cleaning step for noisy window predictions.
- Continuous emotion timeline is not the aim of the paper.

Paper 5. MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations [5]

This paper gives the MELD dataset. Dialogues have many speakers and each utterance has an emotion label. Emotion can change from one turn to the next, so this data fits conversation work.

Drawbacks:

- Label is for the full utterance, not every small part inside it.
- The dataset itself does not give timeline or transition steps.
- Many baselines also use text; our work uses audio for the timeline.

From these papers we can see that most work stops at one label for a segment. Our project takes window-wise predictions and turns them into a continuous timeline with stable parts and transitions.

References

- [1] A Review on Speech Emotion Recognition: A Survey, Recent Advances, Challenges, and the Influence of Noise. ScienceDirect review paper.
- [2] Speech Emotion Recognition Approaches: A Systematic Review. ScienceDirect systematic review paper.
- [3] Speech Emotion Recognition Based on Multi-Feature Speed Rate and Rhythmic Information. ScienceDirect research paper.
- [4] Speech Emotion Recognition Based on Multi-Dimensional Feature Extraction and Multi-Scale Feature Fusion. ScienceDirect research paper.
- [5] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria and R. Mihalcea, MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations, Proceedings of ACL, pp. 527-536, 2019.
- [6] S. Y. Chen, C. C. Hsu, C. C. Kuo and L. W. Ku, EmotionLines: An Emotion Corpus of Multi-Party Conversations, arXiv:1802.08379, 2018.